MIT 18.06SC - 2.2

Problem 11.1

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} +$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Let $c_1P_1 + c_2P_2 + \cdots + c_5P_5 = 0$, where P_i are in the same order as the right hand side of the sum above. Then we have $c_1P_1 + \cdots + c_5P_5 =$

$$\begin{bmatrix} c_2 & c_3+c_5 & c_1+c_4 \\ c_3+c_4 & c_1 & c_2+c_5 \\ c_1+c_5 & c_2+c_4 & c_3 \end{bmatrix}$$

where each entry equals 0. On the diagonal are the entries c_1, c_2, c_3 which are all hence 0, and $c_3 + c_5 = 0$ and $c_3 + c_4 = 0$ thus $c_4 = c_5 = 0$ and so the five matrices are linearly independent.

Problem 11.2

a) ~~A Any~~ Any X in M which has all of its columns as any multiple of $[1 \quad 1 \quad 1]^T$, where the multiple need not be the same for all columns

b) Matrices whose columns are each ~~linear~~ some linear combination of the columns of A. Also, since the columns of A have entries that sum to 0, so should the columns of these matrices.

c) Dim $N(A) = 3$ and dim $C(A) = 6$. Adding them gives the dimension of M i.e 9 ~~and~~